



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

H2 BIOLOGY

Paper 1 Multiple Choice

9744/01

16 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the Multiple Choice Answer Sheet (MCAS).

Write in soft pencil.

Do **NOT** use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

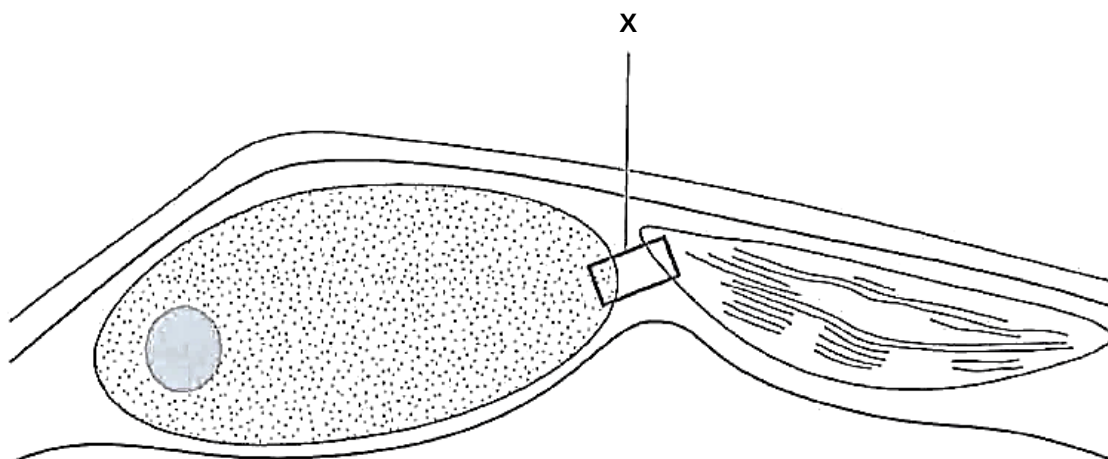
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Answer **all** the questions.

- 1 The diagram shows part of a plant cell, as seen under an electron microscope.



How many layers of membrane are present in the region labelled X?

- A 2
B 3
C 4
D 5
- 2 Two of the isomers of glucose, α -glucose and β -glucose, can be condensed in living cells to form a variety of compounds with properties suited to different functions.

Which row correctly describes a polymer of glucose found in some living organisms?

	molecular structure	isomer	bonds	function
A	branched chains	β -glucose	1,4	energy reserve
B	coiled chains	α -glucose	1,4	structural support
C	branched chains	α -glucose	1,4 and 1,6	energy reserve
D	straight chains	β -glucose	1,6	structural support

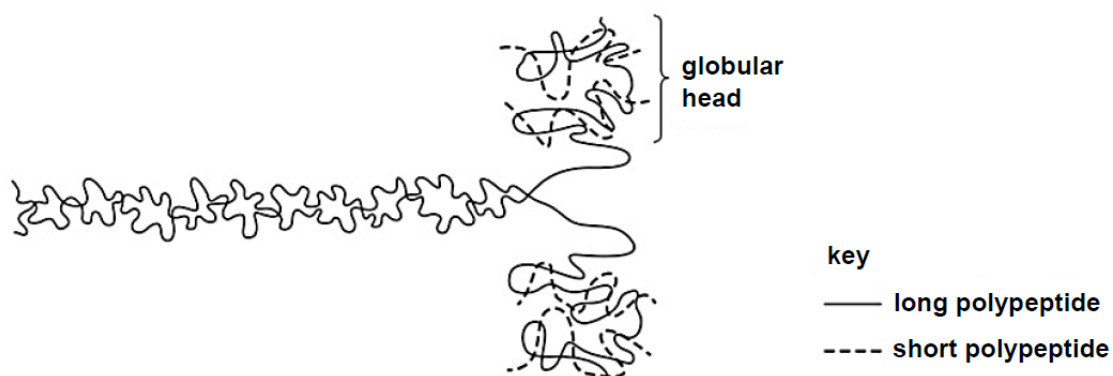
- 3 One way of representing fatty acid chains is $C_x : y$, where C_x is the number of carbon atoms and y is the number of double bonds.

The table shows the percentage mass of eight different fatty acid chains in four cell membranes, **A**, **B**, **C** and **D**.

Which cell membrane would be expected to have the greatest fluidity?

	percentage mass of each fatty acid chain							
	C10 : 0	C12 : 0	C14 : 0	C16 : 0	C18 : 0	C18 : 1	C18 : 2	C18 : 3
A	1	2	4	22	2	52	11	1
B		2	3	23	21	43	4	
C			2	23	19	45	3	1
D	1	2	2	21	35	29	2	1

- 4 The diagram shows a protein molecule.



Two long polypeptides each form α -helices for much of their length and these twist together into a fibre.

At one end, each of these polypeptides coils into a globular head. Two short polypeptides bind to each globular head.

What describes the protein structure of this molecule?

- A** secondary structure because the long polypeptides form α -helices
- B** tertiary structure because the α -helices form a fibre
- C** tertiary structure because the heads form globular proteins
- D** quaternary structure because each molecule consists of six polypeptides

5 How many statements about transport across membranes are correct?

- 1 Facilitated diffusion, osmosis and receptor-mediated endocytosis do not require ATP to proceed.
- 2 In active transport, small non-polar molecules require transport proteins to move across the phospholipid bilayer.
- 3 The secretion of antibodies by plasma B cells into the bloodstream is an example of exocytosis.
- 4 Water molecules are able to move across the phospholipid bilayer directly via osmosis.

A 1

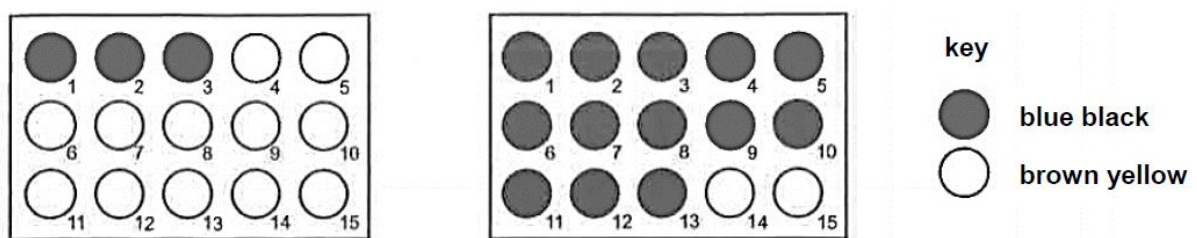
B 2

C 3

D 4

6 An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15-second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution.

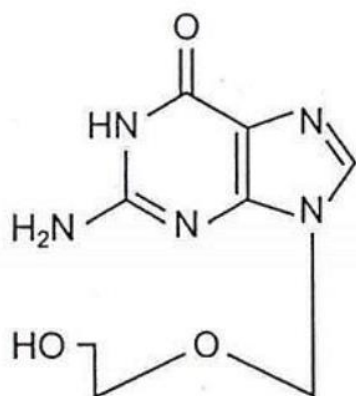
The results are shown in the diagram.



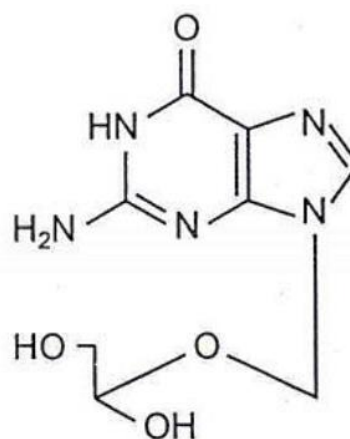
Which shows the correct temperatures and times for the complete digestion of starch?

	temperature / °C	time / s
A	30	3.15
	10	0.45
B	30	45
	10	195
C	10	45
	30	195
D	10	3.15
	30	0.45

- 7 The diagram shows the structures of some drugs that are analogues of viral nucleotides. They interfere with viral nucleic acid synthesis so that incomplete polynucleotides are formed.



acyclovir



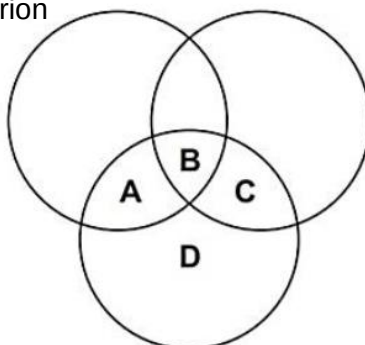
gancyclovir

Which statement explains how these drugs interfere with viral nucleic acid synthesis?

- A They are competitive inhibitors of DNA or RNA polymerase enzymes.
 - B They bind irreversibly to the active site of the viral DNA or RNA polymerase.
 - C They bind to the active site of the host cell DNA or RNA polymerases.
 - D They replace nucleotides so inactive nucleic acids are synthesised.
- 8 Which processes involve mitosis?

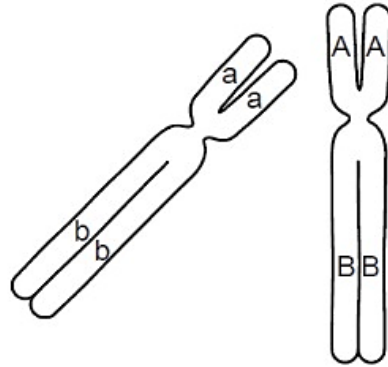
mitochondrial fission
where one mitochondrion
divides into two

forming cells at the tip
of a growing root



replacing skin cells
damaged by an injury

- 9 The diagram shows two homologous chromosomes in early Prophase I of meiosis in an animal cell. Two genes, **A/a** and **B/b**, whose loci occur on the homologous chromosomes, are also shown.



Which row of diagrams is a possible representation of these chromosomes as they progress from anaphase I to prophase II?

	anaphase I	prophase II
A		
B		
C		
D		

10 Which molecules are found in both the cytoplasm and the nucleus of a typical eukaryotic cell?

- 1 DNA nucleotides
- 2 DNA polymerase
- 3 RNA nucleotides
- 4 RNA polymerase

- A** 3 and 4
- B** 1, 2 and 4
- C** 2, 3 and 4
- D** 1, 2, 3 and 4

11 Two students were discussing the involvement of DNA and RNA in transcription and translation.

- Student 1 always stated correct facts.
- Student 2 gave further information, which was sometimes correct.

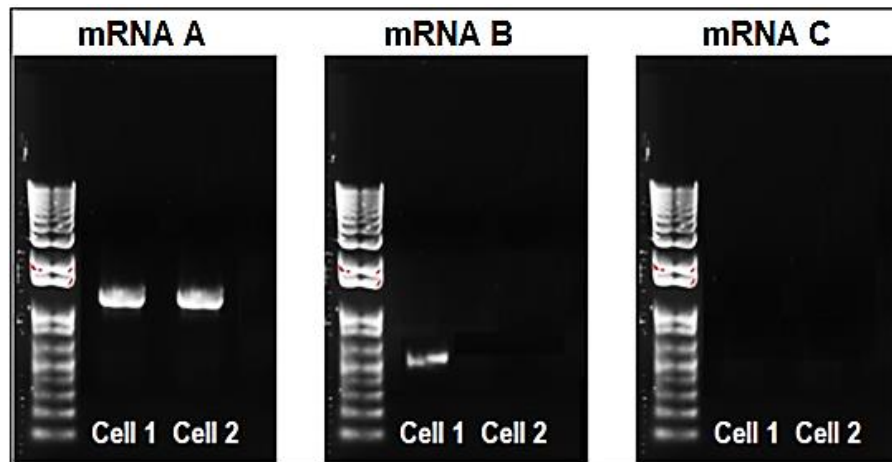
	correct facts given by student 1	further information given by student 2
1	A length of mRNA is 747 nucleotides long, including stop and start codons.	This mRNA can produce a polypeptide that is 249 amino acids long.
2	Adjacent mRNA codons of AAU and CUG bind to complementary tRNA anticodons.	There is a total of 14 hydrogen bonds formed between these two codons and their anticodons.
3	RNA polymerase catalyses the formation of the mRNA from the template strand of DNA.	During translation, an RNA adenine nucleotide will pair with a DNA thymine nucleotide.
4	A DNA adenine nucleotide is structurally different to an RNA adenine molecule.	The difference is in the hexose sugars; DNA is deoxyribose and RNA is ribose.

Which further information, given by student 2, is correct?

- A** 2
- B** 4
- C** 1 and 4
- D** 2 and 3

- 12** Cells were obtained from the islet of Langerhans and the inner cell mass of an embryo. Alpha cells and beta cells from the islet of Langerhans secrete glucagon and insulin respectively. Various mRNAs coding for different proteins were isolated from the cells. Gel electrophoresis was then carried out and specific radioactive probes complementary to three of the mRNAs were added.

The diagram below shows the results.



Which of the following correctly identifies the cells and the proteins coded for by the various mRNAs?

	cell 1	cell 2	mRNA A	mRNA B	mRNA C
A	alpha cell	zygotic stem cell	histone proteins	glucagon	telomerase
B	beta cell	embryonic stem cell	telomerase	insulin	RNA polymerase
C	embryonic stem cell	beta cell	RNA polymerase	telomerase	glucagon
D	zygotic stem cell	beta cell	histone proteins	telomerase	insulin

- 13** Transcription in eukaryotic cells results in the formation of pre-mRNA, which is made up of exons and introns.

Which of the following statements correctly describes what happens during alternative splicing?

- A** All of the introns are removed from the pre-mRNA.
- B** Some of the introns are removed from the pre-mRNA.
- C** Some of the exons are removed from the pre-mRNA.
- D** Some of the introns and exons are removed from the pre-mRNA.

- 14** A single skin cell was isolated and transferred to growth medium in a sterile Petri dish. The Petri dish was incubated for 16 days. During this time, the number of skin cells multiplied as a result of repeated mitotic divisions.

One of the chromosomes in the nucleus of each skin cell has a telomere that contains many repeats of the base sequence TTAGGG. On each of days 4, 8, 12 and 16 of the incubation period, a single cell was removed from the Petri dish. The total number of bases in the telomere of this chromosome was determined for each cell. Each of the four cells had a different total number of telomere bases for this chromosome: 5548, 5580, 5645 and 5700.

What was the total number of telomere bases in the chromosome from the cell that had undergone the most mitotic divisions?

- A** 5548
B 5580
C 5645
D 5700
- 15** One individual, organism P, in a sexually reproducing population of mammals has two forms of a protein in its heart muscle. The normal form is identical to that in most other members of the population, while the variant form differs by one amino acid. The diagram shows the amino acid sequences in part of the normal protein and in the same part of the variant protein.

normal protein
 -glu-lys-lys-gly-glu-

variant protein
 glu-lys-gly-gly-glu

What could account for the presence of the normal protein and the variant protein in P?

- 1 The sequence of amino acids was altered in a cell in the heart muscle of P's mother.
 - 2 The sequence of amino acids was altered in a cell in the ovaries of P's mother.
 - 3 The sequence of bases was altered in the DNA in a cell in the heart muscle of P.
 - 4 The sequence of bases was altered in the DNA in a cell in an ovary of P.
 - 5 The sequence of bases was altered in the DNA in a cell in a testis of P's grandfather.
- A** 1 and 3
B 3 and 5
C 2, 4 and 5
D 1, 2 and 4

- 16** If DNA is damaged, checkpoints in the cell cycle can either trigger DNA repair, allowing the cell to progress through the cell cycle or, if this cannot be carried out, divert the process to programmed cell death (apoptosis).

Breaks in double-stranded DNA can be repaired using proteins such as p53 and Chk1.

About half of all cancer cells have non-functional p53 proteins.

An inhibitor for Chk1 protein has been developed as a treatment for cancer patients to improve tumour shrinkage during radiation treatment.

How would this Chk1 inhibitor benefit these patients?

- A** *Chk1* genes would be damaged and unable to repair DNA.
 - B** Fewer healthy cells would have damaged DNA.
 - C** More cells with non-functional p53 protein would undergo apoptosis.
 - D** The radiation treatment would kill all the tumour cells.
- 17** Which process in bacteria **always** allows chromosomal and non-chromosomal DNA to be transferred?
- A** binary fission
 - B** conjugation
 - C** transduction
 - D** transformation

- 18** In the fruit fly *Drosophila melanogaster* the gene for eye colour is sex-linked. The allele for red eye (R) is dominant to the allele for white eye (r).

A cross between two flies produced the offspring shown in the table below.

sex of offspring	number with white eyes	number with red eyes
female	23	22
male	21	22

The genotypes of the parents in this cross were

- A** X^rX^r and X^{RY}
- B** X^RX^r and X^rY
- C** X^RX^r and X^{RY}
- D** X^RX^R and X^rY
- 19** Seeds from a pure breeding plant were planted in identical pots of compost and watered regularly. Sets of ten pots were placed in different light conditions and left until the first leaves had developed.

The table shows the mean height for the young stems, mean length of first leaf and the colour of the leaves.

	no light	dim light	bright light
mean height / cm	8	6	4
mean leaf length / cm	1.5	1.4	1.4
colour of leaves	pale yellow	pale green	dark green

Which explains the effect of light on the phenotype of the young plants?

- A** The activity of genes involved in chlorophyll synthesis and stem growth varies with light intensity.
- B** The activity of genes involved in stem and leaf growth is decreased by light.
- C** The genes involved in chlorophyll synthesis and stem growth are activated by light.
- D** The genes involved in chlorophyll synthesis and stem growth are inactivated by light.

- 20** The chi-squared (χ^2) test was used to determine whether the inheritance pattern of a dihybrid cross was statistically significant.

Four phenotypes were produced from the cross.

χ^2 was calculated as 8.62 and the significance level (p) was 0.05.

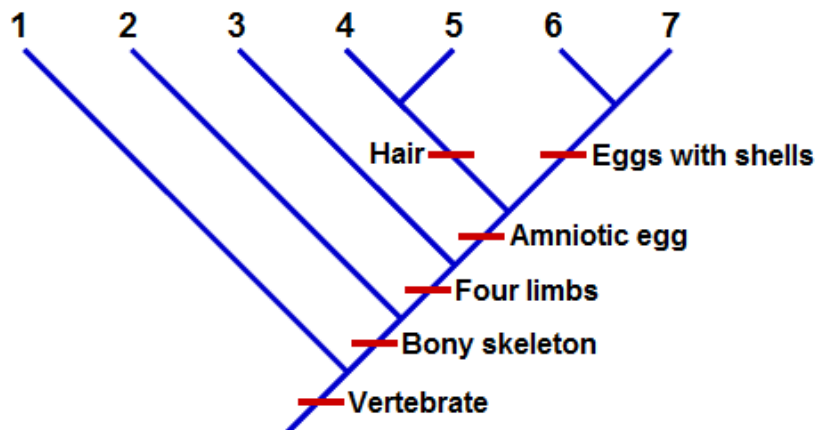
A table of χ^2 values is shown below.

	probability (p)				
df	0.50	0.10	0.05	0.01	0.001
1	0.46	2.71	3.84	6.63	10.83
2	1.39	4.60	5.99	9.21	13.82
3	2.37	6.25	7.81	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46

Which of the statements is correct?

- A** df is 3 and the p -value is more than 0.05, hence the deviation between observed and expected results is not statistically significant.
- B** df is 3 and the p -value is less than 0.05, hence the deviation between observed and expected results is statistically significant.
- C** df is 4 and the p -value is more than 0.05, hence the deviation between observed and expected results is not statistically significant.
- D** df is 4 and the p -value is less than 0.05, hence the deviation between observed and expected results is statistically significant.

21 The following shows the phylogenetic tree of seven animals.

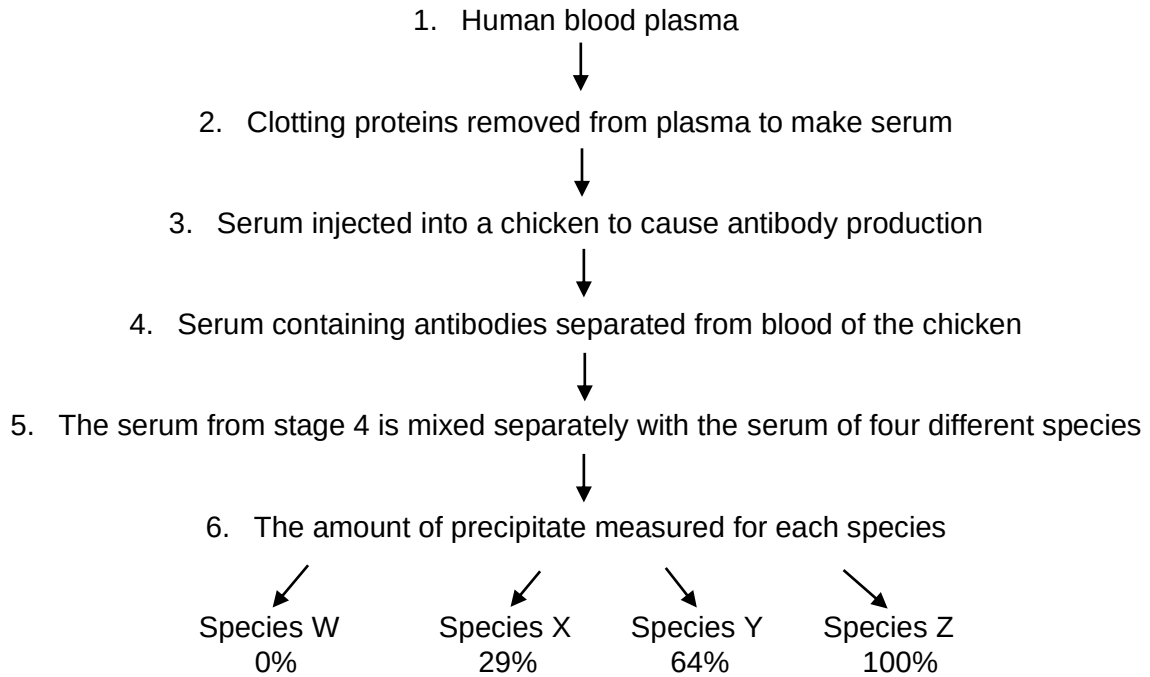


Which statements are true?

- 1 Members of species 3 possess four limbs but not amniotic egg.
- 2 The common ancestor of species 5 and 6 possessed bony skeleton.
- 3 Members of species 4 and 7 both possess amniotic egg due to convergent evolution.
- 4 Species that have hair are most closely related to species that have eggs with shells.

- A** 1 and 2
B 2 and 4
C 1, 2 and 3
D 1, 3 and 4

- 22** The flow diagram shows an investigation into the blood proteins of a number of species. Foreign blood proteins cause the production of antibodies. Mixing an antibody with its corresponding protein causes precipitation.



Which statements are true?

- 1 Species W is human.
 - 2 Species Y is evolutionarily more related to humans than species X.
 - 3 The clotting proteins removed from the blood plasma in step 2 serve as antigens in the chicken.
 - 4 Serum extracted from the blood of the chicken in step 4 are human antibodies.
- A** 1 and 4
- B** 2 and 3
- C** 1, 2 and 3
- D** 2, 3 and 4

23 Which statements regarding signal transduction pathways are true?

- 1 Different cellular responses are possible in a single cell.
- 2 Different cellular responses are possible in different cells.
- 3 Activation of relay molecules always occur.
- 4 Activation of genes always occur.

- A** 1 and 2
B 3 and 4
C 1, 2 and 3
D 1, 2, 3 and 4

24 A student wrote three statements about cell signalling.

- 1 A signal chemical always has the same shape as a protein receptor on a target cell.
- 2 An increase in temperature may decrease the effect of cell to cell signalling.
- 3 A mutation may decrease production of active protein receptors for the cell surface membrane.

Which statement(s) is/are correct?

- A** 1 only
B 1 and 3
C 2 and 3
D 1, 2 and 3

25 Which of these statements are true regarding the process of PCR?

- 1 Although *Taq* DNA polymerase lacks proofreading activity, it is accurate enough to allow PCR to be used for DNA analysis.
- 2 DNA primers used for PCR can be obtained naturally from the target organism.
- 3 The extension step of PCR takes place at about 72°C as that is the optimum temperature of *Taq* DNA polymerase.
- 4 The number of strands of DNA formed after 25 cycles of PCR is 2^{25} .

A 1 and 3

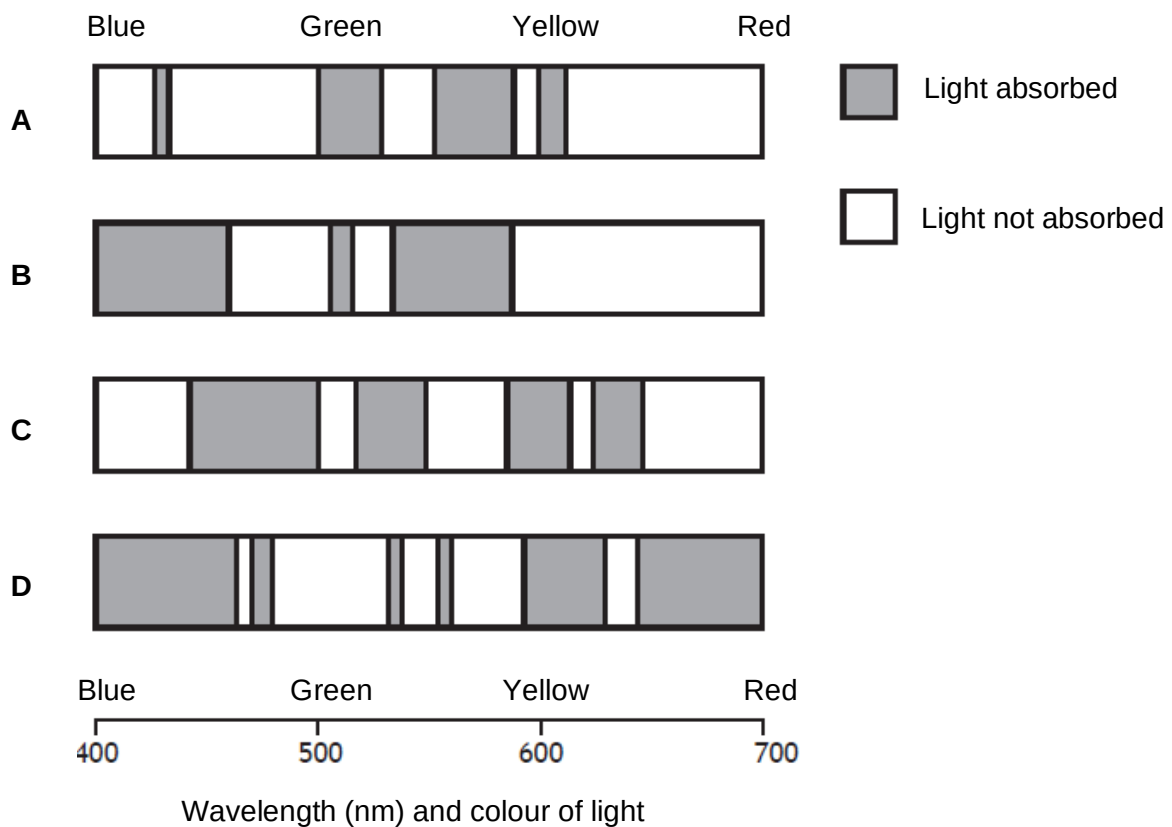
B 1 and 4

C 2 and 3

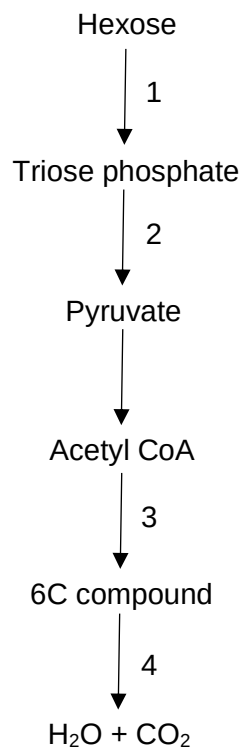
D 2 and 4

26 The following absorption spectra were obtained by testing four different plant extracts.

Which extract contains chlorophyll?



27 The diagram summarises the pathway of glucose breakdown.



Which two steps result in a net increase of ATP?

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

- 28** An antibody, specific for a virus, was treated with an enzyme to break the bonds between the variable and constant regions.

The separated variable and constant regions were then added to cells infected with the virus.

Based on the information provided, which statement(s) is/are correct?

- 1 The constant regions would bind to different parts of the virus antigens.
- 2 The viruses could be engulfed by phagocytes if they were present.
- 3 The variable regions would all bind to the same part of the virus antigens.

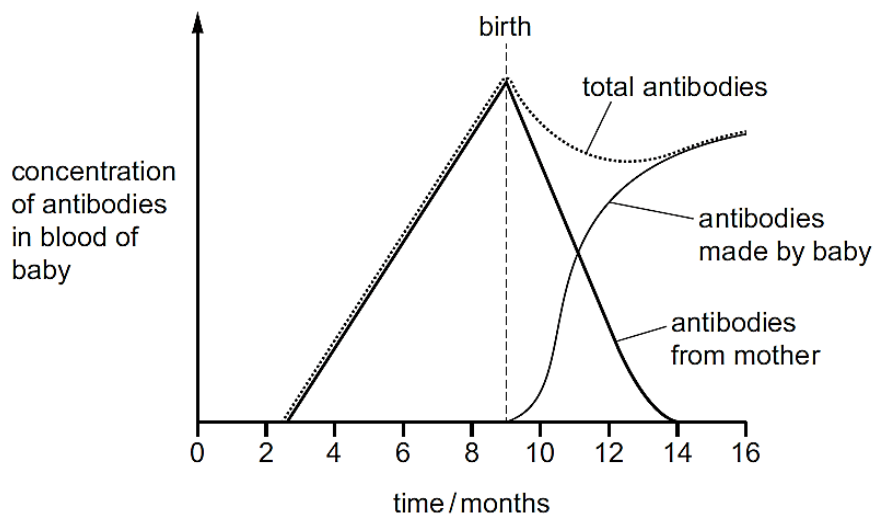
A 2

B 3

C 1 and 2

D 1 and 3

- 29** The graph shows the changes that occur in the concentration of antibodies in the blood of a baby before birth and during the first few months after birth.



Which description about the changes in immunity during the first few months after birth is correct?

- A** active artificial immunity decreases, active natural immunity increases
- B** active natural immunity decreases, active artificial immunity increases
- C** passive artificial immunity decreases, active natural immunity increases
- D** passive natural immunity decreases, active natural immunity increases

- 30** Human activity is believed to have contributed to climate change. It has been suggested that climate change has resulted in a worldwide loss of biodiversity.

Which statements explain why there may have been a loss of biodiversity?

- 1 habitat loss
- 2 invasion from immigrating species that replace native species
- 3 climatic conditions change faster than species can adapt by evolution
- 4 colonisation of previously uninhabited areas

- A** 1 and 3
- B** 2 and 4
- C** 1, 2 and 3
- D** 1, 2 and 4

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